Task: Beyblade Scaling

Background

In order to make it harder for enemy robots to shoot our robot, we have a “beyblade” mode where the Chassis will spin rapidly while the Gimbal remains (relative) stationary.

Currently, this spin rate of the Chassis is the same no matter what. However, for ease of control, what we want to implement is some algorithm that will **Scale the spinning rate down** by the amount our robot is trying to **translate**.

Basically, if we are about to move fast translationally, we want less energy going to spinning and more to translating. And, if we are moving slower (and more likely to be engaging a robot) we want to be spinning faster. This ‘scaling’ will make our rapid movements much more efficient and ultimately faster, helping our robot navigate into and out of engagements better while drawing less power when not needed.

You will implement this by making the the beyblading “scaling” for the offset (Found as a 0-1 DEFINE in in cmd_controller) a runtime variable, linked to the range of the Left Throttle stick. If the throttle is closer to 0, we would want to be spinning fast, if throttle is high, we’d prefer slower spinning (We want spin rate and throttle/translational command to be inversely related). You need to figure out how (and where) you will use the RC frame information to scale the beyblading math

What You Need To Do

Step 1: Understand how the Beyblading Math causes “spinning.” Understand which controller control enables beyblading.

Step 2: Change the scale offset for beyblading to change as the inverse of the throttle command from the RC (think, how will you use the message center to get this info from the RC?)

Step 3: Test this out on the Old Standard. If something goes wrong, debug and fix it.

DO THIS ALL IN A NEW BRANCH OF CODE! TITLE IT SOMETHING APPROPRIATE

Feel free to consult the internet, Other teams’ code, AI, or any other resources to help you debug/strategize how you will implement your algorithm. I highly recommend actually reading through the RC and cmd controller code yourself, and asking leads for help, before trying to AI through it.

Note:Use logger prints when debugging. If you use the raw Debug Printfs, delete them after verifying everything works before you request a merge.